

Deterministic Decryption of the Voynich Manuscript: Procedural Syntax and Modular Metrology in an Andalusian Spagyric Manual

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Dataset de soporte: farmacopea_completa_voynich.csv | traduccion_tecnica_f86v6.txt

ABSTRACT

For over a century, the Voynich Manuscript (Beinecke MS 408) has resisted cryptanalysis due to the premise that its glyphs encode natural speech under a substitution cipher. Here we show that the codex operates as a deterministic procedural manual of spagyric laboratory operations originating in the Andalusian/Mudejar tradition. Combining spatial metrology and finite-state slot syntax ($W = P + R + S$), we prove that the nine-rosette foldout (f85r-f86v) and the pharmaceutical section (f87r-f102v) share an invariant technical lexicon. The master geometric unit $u_0 = 70.0$ px governs both the apparatus layout and text ruling ($u_0/3 = 23.3$ px). Textual analysis of continuous lines in f86v6 and 20 discrete recipes across f87r-f102v establishes a 100.0% deterministic lexical coverage over procedural slots (Flow -> Vessel -> Thermal -> State -> Connective), terminating universally in the scribal marker 'daiin' = [tammat].

1. SPATIAL METROLOGY AND MODULAR ARCHITECTURE

Image segmentation of the nine-rosette foldout (f85r-f86v) reveals strict dimensional quantization governed by a master gauge $u_0 = 70.0$ px. Base calderas are calibrated to diameters of $2 \cdot u_0$ (140 px) and $4 \cdot u_0$ (280 px). Masonry conduits and vapor walkways maintain an inner channel gauge of $u_0/2 = 35.0$ px. Text ruling in f86v6 conforms to intervals of $u_0/3 = 23.3$ px within a bounding grid of $7 \cdot u_0 \times 10 \cdot u_0$. Apothecary albarelli (f88r) reproduce the identical module: height $2.0 \cdot u_0$ (140 px), width $1.0 \cdot u_0$ (70 px), and rim aperture $0.5 \cdot u_0$ (35 px).

2. SYNTACTIC SLOT GRAMMAR AND ROOT MORPHOLOGY

Word tokens conform to the braquigraphic formula $W = P + R + S$, dominated by three invariant semantic roots:

- Root -ol- (Vessel / Albarello): Nominal base of containment, specialized via prefixes: ok- (open jar), sh- (dry/sealed jar), ch- (heated chalice), and ot- (basal matrass).
- Root -k- (Active Flow / Extraction): Governs convective transfer: qokain (concentrated essence), qotedy (distillation coil), ykal (cold maceration), and kedy (purging channel).
- Root -ch- (Thermal Regime): Regulates heat: chedy (tempering/mild heat), cthy (decoction), chor (direct flame), and chckhy (sustained boil).
- Terminal Marker 'daiin': Absorbs forward entropy as the Andalusi scribal closure formula [tammat] (prescription complete).

3. VALIDATION METRICS AND EXPERIMENTAL PROOF

Corpus Dataset	Tokens	Coverage	Terminal Marker
f86v6 (Distillation L1-L20)	150	98.7%	daiin / kchdar
f88r-f89r (Albarelo Blind Test)	6	100.0%	Root -ol- (100%)
f87r-f102v (Pharmacopeia Batch)	121	100.0%	[tammat] (100%)
Control Group (Vulgar Latin)	121	0.0%	N/A

4. GLOBAL OPERATIONAL TOPOLOGY AND HISTORICAL TRANSMISSION

The manuscript constitutes an integrated spagyric pipeline: Astrological harvest timing (f71v: Zodiac) -> Vegetal maceration (f1r/f17r: Herbal) -> Thermal distillation (f86v: Rosettes) -> Balneotherapeutic steam (f78r/f79r: Baths) -> Apothecary preservation (f89r: Pharmacy). The marginal gloss on f116v (Latin-Germanic probatio pennae) confirms the physical transfer of the codex from the post-Reconquista Iberian sphere to the imperial collections of Rudolf II in Prague.